

VARDAN KESHISHYAN

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[LinkedIn](#) | [GitHub](#) | [Portfolio](#)

EDUCATION

University of Pennsylvania

M.S. in Engineering in Artificial Intelligence

Philadelphia, PA

Incoming September 2026

University of California, Irvine

B.S. in Computer Science

Irvine, CA

Completed June 2026

- **Relevant Coursework:** Data Structures, Algorithms, Software Engineering, Database Management, Computer Networks, Artificial Intelligence, Human-Computer Interaction, Systems Design, Discrete Mathematics, Probability, and Statistics

EXPERIENCE

Software Engineer

Bizskip (Unpaid Startup)

Feb. 2026 – May. 2026

Irvine, CA

- Worked with a startup company developing an application that helps users join multiple local business waitlists and find services
- Designed a backend plan for users, **70+** local service providers, service availabilities, waitlists, and notifications
- Fully developed the frontend using React/Next.js, including customer search, provider onboarding, listings, and dashboards
- Integrated Google Places API for business discovery and Twilio SMS for automated waitlist notifications to **200+** users

Research Computing Assistant

Caltech

July 2024 – Sept. 2024

Pasadena, CA

- Resolved **20+** CMS and infrastructure issues while supporting local research systems and data center operations
- Maintained **50+** research-related IT systems and supported reliable data center communication across research operations
- Organized research environments and helped install and configure **15+** servers during the research support period
- Set up and maintained research environments using Ubuntu, Python, and required software tools for technical research work

Data Structures & Linear Algebra Tutor

Learning Center (GCC)

Aug. 2022 – July 2024

Glendale, CA

- Tutored students in Linear Algebra, Discrete Mathematics, Data Structures, and Probability, with **80%** achieving A grades
- Researched and recommended textbooks, software, and supplemental learning materials to improve comprehension
- Developed custom handouts, study materials, and practice quizzes tailored to individual student needs

PROJECTS

LiveLectureAI | *Next.js, TypeScript, Google GenAI Live API, TTS*

- Built a live note-taking web app that listens during lectures and turns the lecture into notes, and summaries with **95%** accuracy
- Synchronized lecture slides with live audio transcription, enabling timestamp-based navigation through **2-hour** recordings
- Added accessibility support with Google Live API, Text-to-Speech, voice help, and lecture review tools for students

Quant Visualizer | *Next.js, TypeScript, FastAPI, Python, Plotly.js*

- Developed pairs-trading analysis platform processing **20+** years of stock data with interactive Plotly.js visualizations
- Implemented correlation, z-score analysis, and backtesting tools for testing simple long-short trading strategies clearly
- Added price trajectories, 3D joint distribution, rolling correlation, Excel reports, and a plain-language AI assistant

RecipeRecommender | *Python, Flask, Scikit-learn, TF-IDF, HTML/CSS*

- Built TF-IDF-based recipe search engine indexing **150,000+** recipes with ingredient-matching algorithm scoring by availability
- Implemented multi-field search (ingredient search, dish-name search, and combined search) with **<100ms** query response time
- Designed responsive Flask interface with filtering by dietary restrictions, prep time, and ingredient substitution suggestions

TECHNICAL SKILLS

Languages: C, C++, Python, Java, R, SQL, JavaScript/TypeScript, HTML/CSS

Frameworks/Libraries: React.js, Next.js, FastAPI, Flask, Tailwind CSS, Google Cloud APIs

Cloud & Databases: MySQL, PostgreSQL, Docker, DigitalOcean, AWS EC2, Supabase

Tools: Git, GitHub, JupyterHub, VS Code, PyCharm, Figma, Vercel, Linux/Unix